GRAVITATION

Gravitational Force

Definition:

Every body in the universe attracts every other body with a force called gravitational force.

Newton’s Law of Gravitation:

The gravitational force between two bodies is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

Formula:

$F = Gm_1m_2 / r^2$

Where:

- F = Gravitational force
- G = Universal gravitational constant
- m_1, m_2 = Masses of bodies
- r = Distance between centers

SI Unit:

Newton (N)

Acceleration Due to Gravity

Definition:

The acceleration produced in a body due to the gravitational force of Earth is called acceleration due to gravity.

Value:

$g = 9.8\ m/s^2$

Formula:

$g = GM / R^2$

Where:

- G = Universal gravitational constant
- M = Mass of Earth
- R = Radius of Earth

Important Points:

- The value of g decreases with altitude.
- The value of g decreases with depth.
- The value of g is maximum at poles.
- The value of g is minimum at the equator.

Weight:

Weight is the gravitational force acting on a body.

Formula:

$W = mg$

Escape Velocity

Definition:

Escape velocity is the minimum velocity required by a body to escape from the gravitational field of Earth.

Formula:

$v_e = \sqrt{2gR}$

Value:

The escape velocity of Earth is approximately 11.2 km/s.

Orbital Velocity

Definition:

Orbital velocity is the minimum velocity required to keep a satellite in its orbit around Earth.

Formula:

$v_0 = \sqrt{gR}$

Value:

Orbital velocity near Earth is approximately 7.9 km/s.

Artificial Satellites:

Artificial satellites are man-made objects revolving around Earth in fixed orbits.

Uses of Satellites:

- Communication
- Weather forecasting
- Navigation
- Remote sensing

Geostationary Satellite

Definition:

A satellite which appears stationary relative to Earth is called a geostationary satellite.

Characteristics:

- Revolves from west to east.
- The time period is 24 hours.
- Moves in circular orbit.
- The orbit lies in the equatorial plane.

Applications of Gravitation:

- Motion of planets around the Sun
- Motion of satellites around Earth
- Falling of objects towards Earth
- Ocean tides due to Moon’s gravity

Quick Revision Points:

- Every body attracts every other body by gravitational force.
- Gravitational force follows inverse square law.
- Acceleration due to gravity is represented by g.
- Weight of body is W = mg.
- The escape velocity of Earth is 11.2 km/s.
- Orbital velocity near Earth is 7.9 km/s.
- Geostationary satellites appear stationary.

Important Formulae:

$$F = Gm_1m_2 / r^2$$

$$g = GM / R^2$$

$$W = mg$$

$$v_e = \sqrt{2gR}$$

$$v_0 = \sqrt{gR}$$